Vaibhav Garg

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EDUCATION

GURU GOBIND SINGH INDRAPRASTHA UNIVERSITY — New Delhi
Dr. Akhilesh Das Gupta Institute of Professional Studies — Delhi
Bachelor of Technology in Computer Science — 2026

PROJECTS

Semantic Video Search & Q&A Engine (RAG Pipeline) | *Python, FastAPI, ChromaDB, Ollama*

- Developed a full stack RAG web application with a FastAPI backend and streaming SSE responses, enabling natural language querying of YouTube course content with precise timestamps and conversation history.
- Built a multi-mode ingestion pipeline with 3 transcription strategies: YouTube subtitle parsing (yt-dlp), local GPU-accelerated Whisper (faster-whisper with auto VRAM management), and Groq Cloud API with retry logic.
- Engineered a semantic search engine using ChromaDB with HNSW cosine-similarity indexing and Ollama nomic-embed-text embeddings, supporting both vector retrieval and natural language timestamp-aware filtering.

FitLog – AI Fitness Tracking Web App | Python, Next.js, TypeScript, PostgreSQL, Django (Link)

- Engineered a full-stack serverless fitness platform using **Next.js 16 (App Router)** with TypeScript, enabling calorie tracking, AI-powered meal logging, and workout analytics, deployed on Vercel with a Python service on Render.
- Built a **cascading LLM fallback pipeline** (Gemini → Groq → OpenRouter) for natural language meal parsing; users type "2 rotis with dal" and the system identifies food items, maps them to a 150+ item nutritional database, and logs macros automatically with sub-3-second response time.
- Designed a **Repository Pattern (DAO) architecture** separating API routes → Service layer → Prisma repositories → Supabase PostgreSQL, ensuring clean separation of concerns and type-safe data access across 10+ database models including meals, workout sessions, weight logs, and weekly insights.
- Integrated **Supabase Auth with Google OAuth** and implemented Edge Middleware (Vercel Edge Runtime) for server-side session validation on every request; protected app routes redirect unauthenticated users in under 5ms without a cold start.
- Applied **Upstash Redis** with sliding-window rate limiting on all AI endpoints and a **TanStack Query** cache layer with optimistic updates, food logged via AI appears instantly in the UI with zero-latency feedback before server confirmation.
- Built a **Django 5.2 + DRF service** on Render sharing the app's PostgreSQL, exposing share-link REST APIs (snapshot model, unguessable slugs, revoke/expiry → HTTP 410, per-user throttling) and a Django Admin for the food catalog; authenticated via a custom class verifying **Supabase ES256 JWTs against cached JWKS**, with pytest trust-boundary tests in CI.
- Built a **workout calorie burn estimator** using MET (Metabolic Equivalent of Task) values and a compound/isolation exercise classification system to calculate burn ranges (e.g., "320–380 kcal") displayed as informational data, never double-counted against the daily budget.

Liver Damage Prediction using Logistic Regression And KNN (Link)

- Integrated Logistic Regression and K-Nearest Neighbors (KNN) algorithms for supervised classification.
- Achieved an accuracy of 86% using the KNN model after hyperparameter tuning and performance evaluation.
- Performed comprehensive data preprocessing, including handling missing values, label encoding, and feature normalization.
- Conducted exploratory data analysis (EDA) to uncover patterns and anomalies in the dataset using Pandas, Seaborn, and Matplotlib.
- Created heatmaps, boxplots, and scatter plots to analyse feature correlations and detect outliers.
- Evaluated model performance using metrics such as accuracy, confusion matrix, precision, recall, and F1-score.
- Used train-test split and cross-validation to ensure the model's robustness and prevent overfitting.

TECHNICAL PROFICIENCIES / SKILLS

CERTIFICATIONS – Verified Credentials

DSA in C++ Coding Blocks (Oct 2023)
Data Science & Machine Learning – IBM (Sept 2024)
Data Analytics – WS Cube Tech (March 2025)
Data Analytics Job Simulation – Deloitte (May 2025)

TECHNOLOGIES

- Languages: Python, TypeScript, C++, SQL
- Backend & Data: Django, Django REST Framework, REST APIs, JWT/OAuth authentication, PostgreSQL, Prisma ORM, Redis, Pandas.
- AI & Machine Learning: Gen AI, RAG pipelines, Embeddings, LLM APIs, Scikit-learn, Regression & Classification Models.